Terrain Development Suitability Model

A multi-layer spatial overlay pipeline that combines soil-derived slope, floodplain, and water service data into a single composite terrain classification, engineered to survive gaps in any one input layer

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The Problem

A land suitability model needs to know, for every parcel and for every square foot of land in the state, which combination of physical constraints applies: is it in a floodplain, does it have water service, and how steep is the terrain. Combining three independently sourced spatial layers into a single composite classification runs into real problems at scale.

- Combining three polygon layers through spatial overlay is one of the most computationally expensive operations in GIS. Doing it directly against millions of parcels can produce tens of millions of tiny sliver polygons, and a naive processing order can make an already expensive operation dramatically worse.
- A composite classification needs a defined precedence rule. What happens when a parcel is both in a floodplain and on a steep slope, which constraint wins.
- Beyond parcels, a separate but related need is measuring the underlying land itself, independent of what's built on it, broken down by the same terrain classes, for higher-level summary and quality checking. That requires the overlay logic to produce consistent, comparable results in a second, independent pass.
- None of the three input layers is guaranteed to fully cover the state. If one layer has a gap and an overlay is built using that layer as its base, it can silently lose land from the total, and the loss doesn't show up as an error, it shows up as missing acreage several steps downstream.
- Any large-scale intersection produces countless tiny sliver polygons just from floating point precision and mismatched boundaries. Left in, they bloat the output and slow every step after. Filtered too aggressively, real small parcels get lost along with them.

My Approach

I built this as a set of five linked ArcGIS Python tools: three data preparation tools that each produce one clean input layer (soil-derived slope class, FEMA floodplain, EPA water service boundary), and two overlay tools, one that tags every parcel with a composite terrain classification, and one that independently measures the underlying land area in the same classes for cross-checking.

Processing order chosen for performance, not convenience. The composite overlay against millions of parcels touches three layers of wildly different sizes, floodplain and water service boundaries are a few thousand polygons each, while slope coverage tiles the entire state. I ordered every intersection step from smallest to largest, joining the small layers in first through a spatial join (fast, since one side is tiny) and only running the expensive full geometric intersection against the layer that has to tile everything. That ordering is the difference between an overlay that finishes in minutes and one that doesn't finish at all.

An explicit, single composite classification rule. Every polygon coming out of the overlay ends up with exactly one composite class built from three yes or no facts, water service, floodplain, and slope band, with floodplain given precedence over slope when both apply, since flood risk is the more binding constraint. That single rule is applied consistently everywhere in the pipeline, rather than left as separate flags a downstream user has to combine themselves.

A land-only version of the same overlay for cross-checking. Separately from the parcel overlay, I run the same three-layer intersection logic against zone boundaries directly, independent of parcel data entirely, so there's a defensible answer for how much land in each zone falls into each terrain class regardless of what's built on it. That gives a second number to check the parcel-level results against.

Choosing a base layer that can't silently lose land. An earlier version of the land-only tool used the floodplain layer as its base for the overlay, which quietly assumed that layer covered the entire state. When that assumption stopped being true, a downstream change had removed placeholder non-flood boundary polygons that had made the layer appear complete, the tool didn't error, it just silently returned zero acres for every terrain class in the state that wasn't flood related. Over twenty million acres vanished from the output with no warning. I rewrote it to use the layer that's actually guaranteed to be complete, the slope layer, as the base, so any gap in the flood or water layers can only ever add zero to a class, never remove real land from the total.

Sliver filtering that doesn't erase real detail. Any overlay this size produces enormous numbers of essentially meaningless polygons where boundaries almost touch. I filter these out by a minimum geodesic area threshold, set low enough that no meaningful real parcel fragment gets dropped, keeping output size manageable without quietly losing legitimate small slivers of land.

Built-in reconciliation checks. After the land-only overlay finishes, it checks its own total against the source zone geometry's area directly, flags any class that comes out statewide zero (the exact failure signature from the earlier bug), and reports the largest per-zone discrepancies, so a regression like the one above gets caught immediately instead of silently propagating.

Technical Highlights

- **Language/Platform:** Python, ArcGIS Pro (`arcpy`) Python Toolboxes and standalone scripts for performance-critical overlay stages

- **Data sources:** County-level soil survey slope classification, FEMA flood hazard boundaries, EPA water service area boundaries, all built earlier in the same pipeline
- **Spatial overlay:** Multi-layer geometric intersection ordered by feature count for performance, spatial join used in place of full intersection wherever one side of a comparison is small
- **Classification:** A single composite terrain suffix built from three binary spatial facts, with an explicit precedence rule
- **Scale handling:** Sliver filtering by minimum geodesic area, in-memory intermediate processing to avoid unnecessary disk writes
- **Data integrity:** A base-layer choice that guarantees full coverage regardless of gaps in supporting layers, with reconciliation checks against independent totals after every run
- **Output:** A fully tagged parcel-level dataset plus an independent, land-area-only version of the same classification for cross-validation

Why It Matters

A composite terrain suitability classification is only as good as the overlay logic underneath it, and the failure mode that matters most in an overlay like this isn't a crash, it's a silent, plausible-looking wrong answer. Choosing a base layer that structurally can't drop land, and building in a reconciliation check that catches a statewide-zero class immediately, meant a real regression that had already shipped and produced twenty million acres of missing land got caught and fixed before it reached a deliverable, rather than being discovered by someone downstream much later.